



Ben Keita

Full Stack Software Developer| Fintech & Wallet Systems |API Development & cloud computing |Team Leadership

Professional Summary

Backend Software Engineer with experience designing, developing, and integrating scalable backend systems for fintech, digital wallets, and payment platforms. Skilled in building secure RESTful APIs, microservices, and cloud-native applications using Python (FastAPI), MongoDB, Docker, AWS, and JWT authentication. Experienced in payment integrations, transaction processing, wallet management, and API communication across distributed systems.

At NBK Diaspo, I develop backend services powering cross-border payments and digital wallet solutions, contributing to secure authentication, transaction workflows, and scalable microservice architectures. Previously at InTouch South Africa, I worked on payment integrations, POS deployments, and the Sanlam Insurance integration project, supporting API communication, transaction validation, production troubleshooting, and end-to-end testing in live financial environments.

Currently expanding my expertise as a Backend AI Engineering Intern at FlyRank, where I am developing practical skills in AI-powered backend systems, Large Language Model integrations, Retrieval-Augmented Generation , vector databases, AI APIs, and scalable AI application development. Passionate about combining backend engineering with artificial intelligence to build intelligent, secure, and high-performance software solutions.

Strong foundation in software architecture, cloud technologies, API design, payment systems, and modern backend engineering, with a passion for solving complex engineering problems and building scalable products for global users.

Core Skills

Backend Engineering	FastAPI, Flask, REST APIs, Microservices, API Design, Backend Architecture, Node.js
Frontend & Mobile	Flutter, React Native, ReactJS basics, Mobile/API integrations
Fintech & Integrations	Wallet services, Payment integrations, Moneytrans, Visa, Sanlam integration Cross –Border-Payments
Databases	MongoDB, MySQL,SQL
Cloud & DevOps	Docker, Git, GitLab,GitHub, Postman, Linux, CI/CD ,Azure, AWS, Apache kafka
Security	JWT Authentication, AWS cognito, API Security
Languages	Python, HTML, CSS, JavaScript, Java, Dart, C++ , C# , Php
AI Engineering	LLM Integration, Prompt Engineering, Retrieval-Augmented Generation, Vector Databases, AI APIs, AI Agent Integration

Professional Experience

Backend AI Engineering Intern | FlyRank AI (*Internship*)

2026 – Present

- Collaborate with software engineers to design, develop, and maintain scalable backend systems powering AI-driven products and intelligent applications.
- Develop and maintain RESTful APIs, backend services, and data pipelines to support Large Language Model (LLM) integrations and AI-powered workflows.
- Contribute to the development of AI Agents and backend orchestration that automate business processes and improve user experiences.
- Participate in integrating AI models into production-ready backend architectures while ensuring performance, scalability, and reliability.
- Support testing, debugging, and optimization of AI services across development and staging environments.
- Work with modern backend engineering practices including API design, cloud-based services, version control, and collaborative Agile development.

Technologies: Python, FastAPI, AI Agents, LLM Integrations, REST APIs, GitHub, Docker, AI Workflows

Backend Software Engineer | NBK Diaspo

2023 – Present

- Design, develop, and maintain scalable backend services supporting digital wallets, cross-border payments, and fintech platforms using Python (FastAPI), MongoDB, Docker, and AWS.
- Build and optimize secure RESTful APIs for wallet management, user authentication, beneficiaries, and financial transaction processing.
- Develop and integrate backend services within microservice-based and distributed architectures, enabling secure communication between platform components.
- Implement secure authentication and authorization using AWS Cognito, JWT, and API security best practices.
- Participate in deployment automation, CI/CD pipelines, containerization, and cloud-based application delivery.
- Collaborate with cross-functional engineering teams to deliver secure, scalable, and high-performance financial solutions.
- Contribute to payment processing workflows, backend integrations, and transaction lifecycle management while maintaining reliability and performance.
- Support debugging, monitoring, and optimization of production services to improve platform stability and user experience.

Technologies: Python, FastAPI, MongoDB, Docker, AWS, JWT, REST APIs, Microservices, CI/CD, Git, Gitlab, Postman, Event-Driven Architecture (Exposure)

Software Developer & Technical Consultant | InTouch South Africa

December 2024 – April 2026

- Supported large-scale deployment and configuration of Android-based POS devices across TotalEnergies service stations throughout South Africa.

- Contributed to the successful implementation of the Sanlam Insurance payment integration within the Total PDA payment ecosystem.
- Integrated backend services and APIs supporting secure payment processing and insurance transaction workflows.
- Investigated production issues by analyzing backend logs, API requests/responses, payment transactions, and commission calculations to identify and resolve discrepancies.
- Validated financial transactions and participated in end-to-end testing to ensure system accuracy, stability, and compliance.
- Collaborated with software engineers, business stakeholders, and external partners throughout deployment, integration, and production support activities.
- Provided technical support during production rollouts while ensuring payment systems operated reliably in live environments.
- Assisted with troubleshooting, software updates, and APK deployment for Android payment terminals.

Technologies: JAVA, Android, Payment Systems, Postman, Linux, API Integration, Transaction Validation, Production Support

Key Projects

Diaspo Platform – Cross-Border Payments & Digital Wallet Ecosystem

Contributed to the backend development of a fintech platform enabling secure cross-border money transfers, digital wallet management, and community savings (Tontines) across multiple African markets. Designed and maintained scalable backend services supporting wallet operations, beneficiary management, authentication, and financial transaction processing. Collaborated within a microservices-based architecture to ensure secure API communication, scalability, and high system reliability.

Key Contributions

- Developed secure RESTful APIs using FastAPI.
- Implemented wallet account and beneficiary management services.
- Integrated authentication using AWS Cognito and JWT.
- Supported transaction processing and backend service communication.
- Participated in Docker-based deployments and CI/CD workflows.

Wallet Services API Platform

Designed, developed, and maintained backend APIs supporting the core functionality of a digital wallet platform. Built secure services for user authentication, wallet management, beneficiary operations, and financial transaction workflows while following modern API design principles and backend security best practices.

Key Contributions

- Designed scalable RESTful API endpoints.
- Developed secure authentication and authorization services.
- Managed wallet accounts, beneficiaries, and transaction records.
- Optimized backend communication between distributed services.
- Supported API testing, debugging, and production maintenance.

Sanlam Insurance Payment Integration

Contributed to the successful integration of Sanlam Insurance services into the InTouch Total PDA payment platform, enabling insurance-related transactions through Android payment devices. Worked closely with engineering and operations teams to ensure secure API communication, transaction validation, and reliable payment processing within production environments.

Key Contributions

- Integrated backend APIs supporting insurance payment services.
- Validated transaction requests and payment responses.
- Investigated backend logs and API payloads to resolve production issues.
- Performed end-to-end testing across live payment environments.
- Collaborated with cross-functional teams during deployment and production support.

POS Deployment & Payment Infrastructure

Supported the nationwide deployment and operational rollout of Android-based POS devices across TotalEnergies service stations in South Africa. Assisted with device configuration, payment application deployment, transaction validation, and production support to ensure secure and reliable payment operations.

Key Contributions

- Installed and configured Android payment applications (APKs).
- Supported payment workflow validation before production rollout.
- Assisted with POS deployment and launch readiness.
- Diagnosed and resolved payment-related technical issues.
- Worked alongside engineering and operations teams during live deployments.

AI Backend Engineering & Intelligent Workflow Automation (*FlyRank AI*)

Contributing to the development of AI-powered backend systems that integrate Large Language Models and AI Agents into scalable production applications. Supporting the design of intelligent backend services that automate workflows, improve system efficiency, and enable AI-driven features through modern backend engineering practices.

Key Contributions

- Develop backend APIs supporting AI-powered applications.
- Integrate Large Language Models (LLMs) into backend services.
- Build AI Agents to automate business processes and user workflows.
- Design scalable AI workflows connecting backend services with AI models.
- Participate in testing, optimization, and deployment of AI-enabled backend systems.

Education

Advanced Diploma in Application Development (*In Progress*)

The Independent Institute of Education (IIE) – Rosebank College, South Africa
Expected Completion: 2026

Relevant Coursework: Backend Development, Software Engineering, Cloud Computing, Database Systems, API Development, System Design, Software Architecture.

Diploma in Information Technology (completed)

Damelin, South Africa
2022 – 2025

Focused on software development, databases, object-oriented programming, networking, systems analysis, and web application development.

Bachelor of Science in Accounting (completed)

Université Félix Houphouët-Boigny, Côte d'Ivoire
2015 – 2018

Developed strong analytical, financial, and problem-solving skills that complement software engineering in fintech and financial technology environments.

Professional Training

Backend AI Engineering Intern (*Current*)

FlyRank AI | 2026 – Present

- Developing AI-powered backend applications using Python and modern software engineering practices.
- Building and integrating AI Agents and Large Language Models (LLMs) into backend services.
- Designing scalable APIs and intelligent workflow automation.
- Collaborating with engineers on backend architecture, testing, debugging, and production-ready AI systems.
- Gaining practical experience in modern AI engineering, cloud-native backend development, and software integration.

Language

- ☐ **French** — Native Proficiency
- ☐ **English** — Fluent (Professional Working Proficiency)

Key Highlights

- ✓ Delivered backend solutions supporting digital wallets, cross-border payments, and fintech platforms.
- ✓ Contributed to secure API development using Python (FastAPI), MongoDB, Docker, AWS, and JWT Authentication.
- ✓ Supported enterprise payment integrations, including Sanlam Insurance, MoneyTrans, and Visa-related payment ecosystems.
- ✓ Participated in the deployment of payment infrastructure across TotalEnergies service stations in South Africa.
- ✓ Developing expertise in AI Engineering, focusing on AI Agents, Large Language Model (LLM) integrations, and AI-powered backend workflows.
- ✓ Experienced working in Agile environments, collaborating with cross-functional engineering teams to deliver secure, scalable, and production-ready software.